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CHARACTERISTICS OF LIVING THINGS

24/08/22

M - Movement

R - Respiration

S - Sensitivity

G - Growth

R - Reproduction

E - Excretion

N - Nutrition

Movement:

- action by all or part of an organism
- causes change in position or place

Respiration:

- chemical reactions in cells
- nutrient molecules broken down
- energy released for metabolism

Sensitivity:

- detecting and responding to changes in the environment
- sensing stimuli in the internal or external environment

Growth:

- permanent increase in size
- increase in dry mass

Reproduction:

- processes that make more of the same kind of organism

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Excretion::

- Removal of toxic materials and substances in excess of requirements.

Nutrition::

- taking in materials for energy, growth and development

[Metabolism]::

- chemical reactions in cells (including respiration)

CLASSIFICATION

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binomial system:

- names organism using both genus and species.

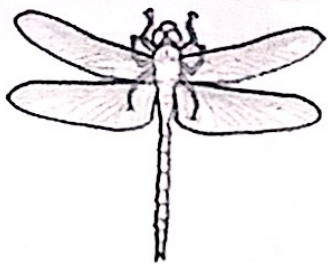
Species:

- groups of organisms that can reproduce to produce fertile offspring.

traditional classification systems grouped organisms based on their morphology and anatomy.

modern classification systems look at the sequences of bases which dictate the order of amino acids.

Dichotomous Keys:



Dragon fly

1	a. More than 6 legs	Arachnid
	b. 6 legs or less	Go to step 2 ✓
2	a. Has wings	Go to step 3 ✓
	b. No wings	Ant
3	a. Single pair of wings	Fruit fly
	b. Two pair of wings	Go to step 4 ✓
4	a. Wings are nearly equal size	Dragonfly ✓
	b. Forewings larger than hindwings	Mayfly

- it is used to identify different organisms based on the organisms observable trait.

groups of organisms which share a more recent ancestor, have more similar base sequences in DNA than those that share only a distant ancestor

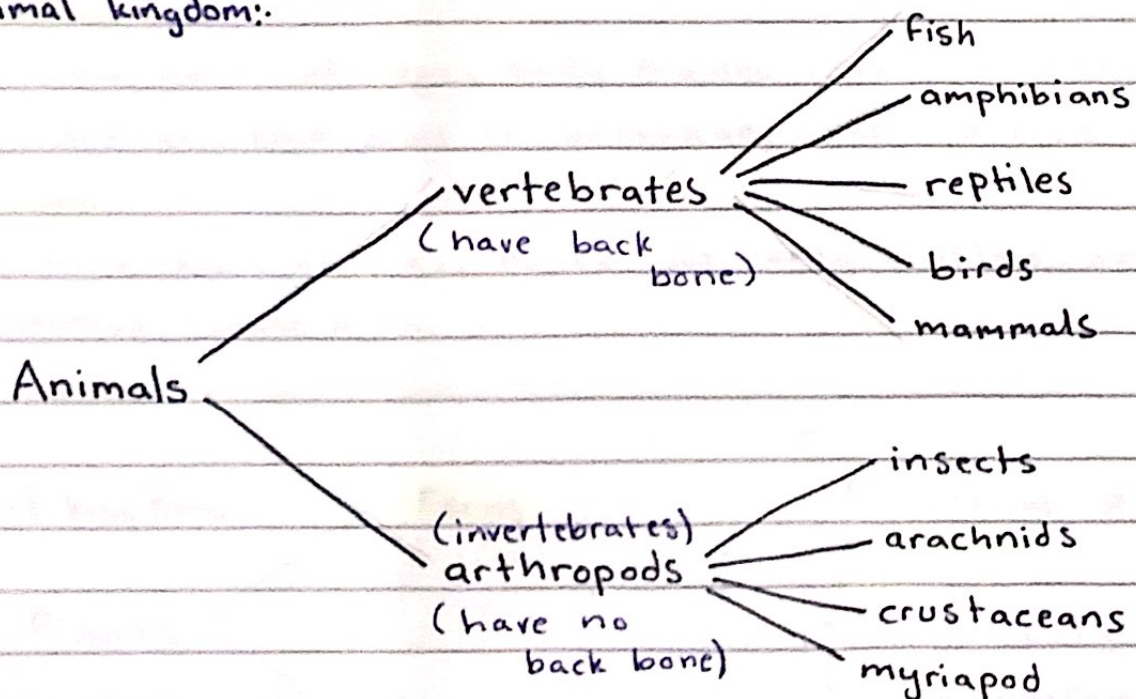
FEATURES OF ORGANISMS

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The 5 kingdoms:

- animals
- plants
- protocists
- bacteria
- fungi

Animal kingdom:



vertebrates:-

Fish:

Scales, fins, eggs in water, eyes

amphibians:

moist skin, 4 limbs (webbed back feet), eggs in water, eyes and ears

reptiles:

dry skin and scales, 4 legs (except snake), shelled eggs on land, eyes and ears

birds:

feathers, wings, 2 legs, eggs with hard shell, eyes and ears

mammals:

Fur, 4 limbs, live young, eyes and ears

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arthropods:-

insects:

3 pair of legs, body divided into head, thorax and abdomen, one pair of antennae, pair of compound eyes, may have wings (usually 2 pairs).

arachnids:

4 pairs of legs, body divided into cephalothorax and abdomen, several pairs of simple eyes. (3)

crustaceans:

5 or more pairs of legs, body divided into cephalothorax and abdomen, two pairs of antennae, compound eyes. (4)

myriapods:

10 or more pairs of legs, no obvious body division, one pair of antennae, simple eyes. (4)

Plant kingdom:-

Plants

Ferns

Flowering plants

Flowering plant:-

monocot:

- single cotyledon
- long, narrow leaf
- parallel veins
- vascular bundles scattered
- floral parts in $\times 3$

dicot:

- two cotyledon
- broad leaves
- network of veins
- vascular bundle in a ring
- floral part in multiples of $\times 4, \times 5$

	Ferns	Flowering plants
pollen	absent	present
seeds	absent	in fruits
Flower / fruit	absent	present
water required for fertilization	Yes	No

Kingdom:

Animal

- eukaryote
- no cell wall
- multi-cellular
- heterotroph

Plant

- eukaryote
- cell wall (with cellulose)
- multi-cellular
- autotroph

Protoctist

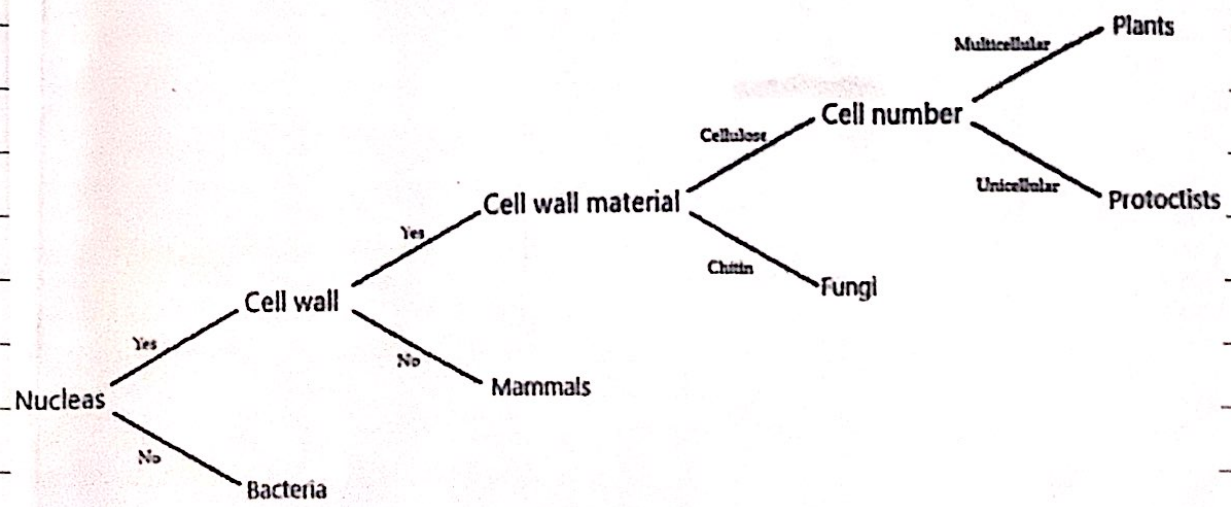
- eukaryote
- cell wall (with cellulose)
- mostly unicellular
- autotroph or heterotroph

Bacteria

- prokaryote
- cell wall (with peptidoglycan)
- unicellular
- autotroph or heterotroph

Fungi

- eukaryote
- cell wall (with chitin)
- multicellular
- heterotroph



-• eukaryote

organisms whose cell contain a nucleus

-• prokaryote

organisms whose cell don't contain a nucleus

-• heterotroph

organism that eats other plants or animals for energy and nutrients

-• autotroph

organism that can produce its own food (using light, water, or carbon dioxide, etc.)

Flowering plant:

- single cotyledon
- long, narrow leaf
- parallel veins
- vascular bundles scattered
- flower buds in axils

Flowering plants

Form

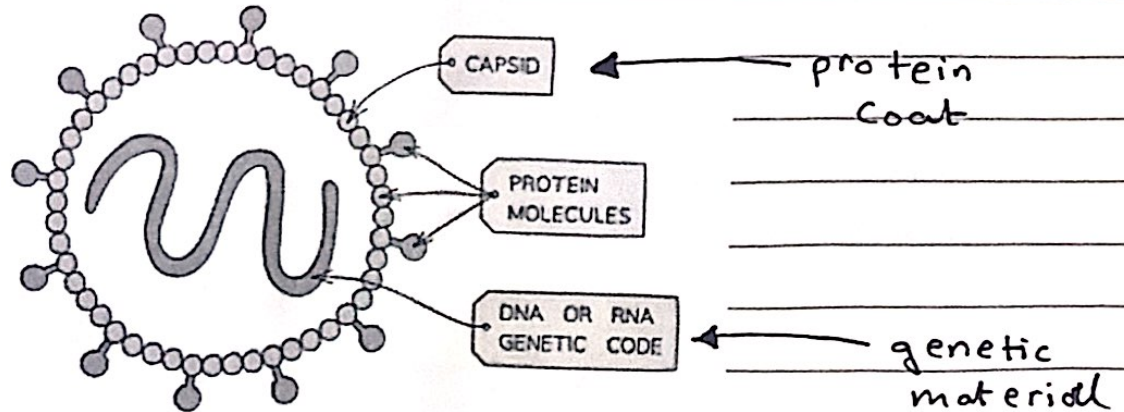
Plant Kingdom

Plants

Flowering plants	Form	pollen	
vascular bundles	scattered	seeds	
long, narrow leaf	parallel veins	flower buds in axils	
single cotyledon			

Virus:

- an infectious agent that can only replicate within a host organism.



- very small (100 times smaller than bacteria)
- only characteristic of life is reproduction
- unusual cell structure